

Joonho Bae

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Research Interests

- Methodology: optimization; dynamic programming; data-driven decision making
- Application: inventory (energy storage) control; energy; system monitoring; recommendation systems

Education

- University of Michigan-Ann Arbor** Sep. 2019 - present
- PhD Candidate in Technology & Operations, Stephen M. Ross School of Business
- Korea Advanced Institute of Science and Technology (KAIST), South Korea** Feb. 2017 - Feb. 2019
- MS in Industrial & Systems Engineering
 - Dissertation: "Frequency-based Change Detection via Multi-Output Log-Gaussian Cox Processes"
 - Committee: Jinkyoo Park (Chair), Il-Chul Moon, and Kyoung-Kuk Kim
- Seoul National University (SNU), South Korea** Mar. 2009 - Feb. 2016
- BS in Statistics and Financial Economics

Publication

1. **Bae, J.** and Park, J. "Count-based Change-Point Detection via Multi-Output Log-Gaussian Cox Processes," *IJSE Transactions*, 52(9), 998-1013, 2020, <https://doi.org/10.1080/24725854.2019.1676937>
- Preliminary version appeared at 2018 *Stochastic Processes and their Applications*, Gothenburg, Sweden (poster presentation)

Works In Progress

2. **Bae, J.**, Kapuscinski, R., and Silberholz, J. "Cost-Saving Synergy: Energy Stacking in Battery Energy Storage Systems," *submitted*.
3. **Bae, J.**, Park, J., Choi, J., and Soh, S. "A Recommending System for Mobile Games using Dynamic Nonparametric Model," *submitted*.
4. Sim, H., **Bae, J.**, and Park, J. "Optimal Management of Energy Storage Systems for Wind Turbines," *in preparation to submit to Renewable Energy*
5. **Bae, J.**, Kapuscinski, R., and Silberholz, J. "Dynamic Valuation of A Battery Under Performance Degradation," *work in progress*.
6. **Bae, J.**, Kapuscinski, R., and Silberholz, J. "Joint Valuation of A Battery Portfolio: Benefits of Heterogeneity," *work in progress*.

Research Experience

- Research Fellow, KAIST** (Advisor: Prof. J. Park, Sponsors: **LG CNS & LG Chem**) 2019
- Developed and transferred real-time system monitoring system exploiting correlations between multi-sensors within energy storage systems (ESS)
 - Implemented a data-driven, nonparametric approach (Gaussian process regression) to estimate state of health (SOH) for ESS
- Research Assistant, Seoul National University** (Advisor: Prof. Sinsup Cho, **Time Series Lab.**) 2012 - 2013
- Revised the Korean Educational Statistics Software (KESS) to enable statistical analysis on Excel

Work Experience

- Marketing Manager, SK Telecom** 2015 - 2017
- Managed more than 40 stores in Seoul
 - Developed optimizing tools for distribution of the cell phones to maximize the profits
- Co-Founder & Project Leader, Cham Son Gil Cooperative** 2013 - 2015
- Established a healing center for blind masseurs and developed a unique B2B service and products targeting 20-30s, who were not the main customers of the original market
 - Expanded branches to nationwide (8 branches as of 2022), ensuring \$3,000 monthly income for each masseur on average
- Sergeant, 8th U.S. Army** 2010 - 2012
- Served in the U.S. Army as a Korean Augmentation to the United States Army (KATUSA)

Teaching Experience

Stephen M. Ross School of Business

- Instructor, Operations Management (TO 313, BBA Core; class size 79)** Fall 2021
- Course evaluation: 4.9/5.0 (Good/Excellent Rating: 100%)
 - Taught twice a week; held weekly office hours; co-developed exams and course materials with other instructors; guided team projects; designed and implemented in-class activities.
 - Topic covered: process analysis, waiting time analysis, inventory management, quality management, demand analytics, project management, supply chain management, and revenue management.

Assistant, MBA Courses

- Manufacturing and Supply Operations (TO 605; class size 80) Fall 2020, 2022, Winter 2020, 2022
- Applied Business Statistics (WMBA 503; class size 114) Summer 2022
- Applied Analytics for Decision Making (WMBA 623; class size 33) Fall 2022

Korea Advanced Institute of Science and Technology (KAIST)

Assistant, BBA Courses

- Data-Driven Decision Making and Control (IE 481; class size 26) Fall 2018
- Engineering Statistics I (IE 241; class size 50) Spring 2018

Conference Presentations & Invited Talks

1. "Dynamic Valuation of A Battery Under Performance Degradation," *Annual POMS Conference 2023*, Orlando, FL, U.S., May 21-25, 2023 **(scheduled)**
2. "Dynamic Valuation of A Battery Under Performance Degradation," *Early-career Sustainable Operations Workshop*, Amherst, MA, U.S., February 25-26, 2023 **(scheduled)**
3. "Cost-saving Synergy: Demystifying Energy Stacking with Battery Energy Storage Systems," *POMS-HK Conference 2023*, Hong Kong, January 7-8, 2023 **(scheduled)**
4. "Dynamic Valuation of A Heterogeneous Portfolio," *INFORMS Annual Conference 2022*, Indianapolis, IN, U.S., October 16-19, 2022
5. "Cost-saving Synergy: Demystifying Energy Stacking with Battery Energy Storage Systems," *MSOM Conference*, Munich, Germany, June 27, 2022
6. "Cost-saving Synergy: Demystifying Energy Stacking with Battery Energy Storage Systems," *INFORMS Annual Meeting 2021*, Anaheim, CA, U.S., October 24-25, 2021
7. "The Recommender System for Mobile Applications," *2018 Global Marketing Conference*, Tokyo, Japan, July 26-29, 2018
8. "Frequency-based Anomaly Detection via Multi-Output Log Gaussian Cox Processes," *2018 Stochastic Processes and their Applications*, Gothenburg, Sweden, June 11-15, 2018.

9. "An Application of Doubly Stochastic Poisson Process for Detecting Abnormalities," *INFORMS Annual Meeting 2018*, Phoenix, AZ, U.S., November 4-7, 2018
10. "The Recommender System for Mobile Applications," Department of Business Graduate Seminar, Yonsei University, South Korea, May 29, 2018

Awards & Honors

- University of Michigan Graduate Teacher Certificate (2022)
- Michigan Ross Conference Travel Grant (2021-2022)
- Michigan Ross Early PhD Candidacy Award (2021)
- National Science & Technology Scholarship, KOSAF (2009-2015)
- Social Venture Idea (\$3,000), Ministry of Employment & Labor, South Korea (2014)
- Enactus National Competition, Enactus (2014)
- Hope Advertisement, Seoul Metropolitan Government (2014)
- SCH Social Venture Idea (\$1,000), Soonchunhyang University (2013)
- SK Social Enterprise (\$10,000), SK Happiness Foundation (2013)
- Army Commendation Medal, 8th U.S. ARMY (2012)

Service Activities

- Session Chair (SB38; Mitigating Climate Risk in the Energy Sector), INFORMS Annual Meeting 2021
- Ross PhD Graduate Student Research Committee Chair (decide on research grants) 2022-2023
- Ross PhD Graduate Student Research Committee Member 2020-2022
- Non-Profit Business Consulting (for the blind masseurs at Cham Son Gil Cooperative) 2015-present

Skills and Activities

- Optimization Tools: Gurobi, CVX
- Programming: C, Python (ML modules: GPy, gpflow, tensorflow), MATLAB, R, SAS
- Languages: Korean (native), English (fluent)
- Hobbies: weightlifting, soccer

References

Roman Kapuscinski (PhD Advisor)

Professor

Department of Technology & Operations

Stephen M. Ross School of Business

University of Michigan, Ann Arbor, MI, USA

E-mail: kapuscin@umich.edu

Jinkyoo Park (MS Advisor)

Assistant Professor

Department of Industrial & Systems Engineering

KAIST, Daejeon, South Korea

E-mail: jinkyoo.park@kaist.ac.kr

Jeonghye Choi (Research Co-Advisor)

Professor

Department of Marketing

Yonsei University, Seoul, South Korea

E-mail: jeonghye@yonsei.ac.kr

John Silberholz (PhD Advisor)

Assistant Professor

Department of Technology & Operations

Stephen M. Ross School of Business

University of Michigan, Ann Arbor, MI, USA

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Seung Bum Soh (Co-Researcher)

Assistant Professor

Department of Operations, Decisions and Information

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